

Tingjian (Terence) Yu

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Profile

Seasoned software engineer with over 13 years of working experience. Expert in designing and developing scalable, high-performance applications serving millions of clients. Adept at delivering clean, testable code and collaborating with cross-functional teams to drive project success.

Work Experience

Senior Software Engineer

Dec 2017 – Mar 2023

Ecovacs Robotics, China

- Led a group of 8 to build over a dozen backend applications, including NLP systems, IoT platforms, image processing, data pipeline, and integration with Alexa and Google Assistant, supporting over 10 million users and devices.
- Designed key architectural solutions focusing on distributed microservices, robust API design, and relational modeling, ensuring scalability, security, and 99.99% high availability.
- Created service-oriented applications with Node.js, Go, Protocol Buffers, REST APIs, PostgreSQL, MongoDB, Kafka, and Redis, handling peak concurrent requests of around 30,000 per second.
- Integrated third-party AI interfaces and built a labeling system to train specialized business models based on the general models of cloud vendors, which increased the speech recognition rate by 7%.
- Optimized legacy product performance through refactoring from monolithic to microservices architecture, rewriting them in compiled rather than interpreted language, reducing interface latency by up to 90%.
- Streamlined software release processes by utilizing tools like Git, Jenkins/GitHub Actions, SonarQube, and automated testing, ensuring efficient and error-free deployments, shortening deployment time by 2/3, and reducing errors by 70%.
- Collaborated closely with cross-functional teams to identify and prioritize work items, delivering incremental software releases in 2-week sprints.
- Mentored a team of 5 junior engineers, improving team effectiveness through pair coding, code reviews, and technology sharing.

iOS Developer

May 2012 – Dec 2017

Ecovacs Robotics, China

- Participated in the definition of mobile application prototypes and the design of UX and UI.
- Built the company's first robot-controlling mobile application from scratch.
- Developed iOS applications with OC and Swift, using Cordova to implement hybrid development.
- Encapsulated IoT communicating SDK, providing abstract interfaces to upper-level developers.

Embedded Software Developer

Mar 2010 – May 2012

Qisda Corporation, China

- Built driver programs for Scanner's optical sensor and motor, manipulating registers to implement functions according to the chip's spec with C and Assembly language.
- Optimized firmware of Printers.

Skills

Languages: Go, JavaScript, Python, Java, C, C#, Objective-C, Swift, Assembler, HTML.

Backend Development: Network Programming, Microservices Architecture, Distributed System, Design Patterns, REST, GraphQL, OAuth, SQL, NoSQL, Cache, Message Queues.

Cloud and Tools: OpenAPI, Git, Agile, AWS, GCP, Docker, Kubernetes, CI/CD, Shell Script, Linux, Elastic Stack, Prometheus.

Education

Postgraduate Certificate of Web Development (Distinction) | *Conestoga College*

2023 – 2024

Bachelor of Electronic Information Engineering (Academic Scholarship) | *Chongqing University*

2005 – 2009

Work Projects as a Senior Software Engineer at Ecovacs

Machine Behavior Controlling based on Natural Language Processing

June 2021 – Apr 2022

- Users needed to control their machines via voice interaction. My responsibilities included the design of the entire software system and the implementation of key codes.
 - Defined flowchart, database schema, Protobuf, and REST API.
 - Implemented a scalable backend architecture using Golang and gRPC to transmit encoded audio streams to third-party NLP platforms like Google Dialog Flow and Azure Speech-to-Text.
 - Decoupled third-party platforms' dependencies by creating modular integrations that can be easily replaced. Leveraged Google Cloud, Kafka, Redis, PostgreSQL, and S3 for efficient audio data processing and system scalability.
- Reached millions of users, with tens of millions of daily interactions three months after launch.

Machine Clean Logs Generating

June 2020 – Dec 2020

- Users experienced serious latency when viewing clean logs on their mobile applications due to CPU-intensive operations running on Node.js based on single-thread and interpreting-running. My duty was to refactor the old project.
 - Rewrote the picture generation program using Golang and OpenCV. Compiled it into the ARM architecture library, which the machine loads as a dynamic library to offload server computation.
 - Employed technologies such as Goroutine, CGO, MongoDB, Kafka, and S3 to enhance data management and streaming capabilities.
- Decreased the log generation time from 1 second to 100 milliseconds, significantly improving performance.

Robot Open Platform and Smart Speaker Integration

Mar 2019 – Sept 2019

- Third-party platforms demanded access to our customers' machines through their own applications. My task was to design a general platform and complete most of the coding work.
 - Designed multiple microservices, such as permission control and instruction scheduling, based on the single responsibility principle.
 - Developed a secure website and standardized APIs enabling third-party developers to register applications, set permissions, and execute operations via JWT and REST APIs after administrator approval, using Node.js, VUE, AWS Lambda, MongoDB, Oauth2.0, etc.
 - Implemented smart home applications based on Alexa and Google Home specifications to interact between speakers and customers' machines.
- Handled millions of daily API calls and thousands of concurrent requests per second at peak times.